

Test Strategy for E-commerce Website

Version 1.0

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Objective

This Test Strategy defines the testing approach, objectives, scope, testing levels, test types, resources, risks, deliverables, and entry/exit criteria for validating the SauceDemo web application. The goal is to ensure the application meets business and user requirements while maintaining high quality and reliability before release.

1. Project Overview

Demo is a sample e-commerce web application that allows users to:

- Log in using valid credentials
- View product inventory
- Sort products
- Add and remove products from the shopping cart
- Complete the checkout process
- Log out successfully

The application serves as a platform to validate end-to-end user journeys and critical business functionalities.

2. Testing Objectives

The primary objectives of testing are:

- Verify that all business-critical functionalities work as expected.
- Ensure users can successfully complete the purchase workflow.
- Identify and report defects before release.
- Validate application behavior under positive and negative scenarios.
- Confirm that changes do not adversely impact existing functionality.

3. Scope of Testing

3.1 In Scope

Authentication Module

- Valid login
- Invalid login
- Locked user login
- Session handling
- Logout functionality

Product Inventory Module

- Product display
- Product details
- Product sorting
- Product pricing validation

Shopping Cart Module

- Add to cart
- Remove from cart
- Cart item persistence
- Quantity validation

Checkout Module

- Customer information entry
- Checkout overview
- Order confirmation

Navigation

- Hamburger menu
- Back navigation
- Cross-page navigation

User Interface Validation

- Button functionality
- Labels and messages
- Field validations
- Error handling

Cross Browser Validation

- IE
- Chrome
- Firefox

3.2 Out of Scope:

The following testing activities are excluded:

- Performance Testing
- Load Testing
- Stress Testing
- Security/Penetration Testing
- Accessibility Testing
- API Testing
- Database Validation

4. Test Levels

4.1 Smoke Testing

High-level validation to verify:

- Application launch
- Login functionality
- Product page accessibility
- Cart functionality
- Checkout completion

4.2 Functional Testing

Validation of business requirements and user workflows.

4.3 Regression Testing

Verification that existing functionality continues to work after changes or bug fixes using Tosca Automation Suite

4.4 Exploratory Testing

Unscripted testing to identify unexpected behaviors and usability issues.

5. Test Types

5.1 Functional Testing

Validate all business requirements.

Examples:

- Login authentication
- Product sorting
- Cart operations
- Checkout completion

5.2 UI Testing

Validate:

- Layout consistency
- Label accuracy
- Button functionality
- Error messages

5.3 Negative Testing

Validate application behavior when invalid data is entered.

Examples:

- Blank username
- Blank password
- Invalid credentials
- Missing checkout information

5.4 Boundary Value Testing

Examples:

- Minimum field lengths
- Maximum field lengths
- Special characters
- Empty values

5.5 Risk Based Testing

Examples

- Login
- Logout
- Checkout Information
- Checkout Completion

5.6 Usability Testing

Validate:

- Ease of navigation
- Clarity of messages
- User experience

6. Test Environment

Application URL

<https://www.saucedemo.com>

Test Environment

QA / Demo Environment

Supported Browser Testing

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

Operating Systems

- Windows 11
- macOS

7. Test Data

Valid User

Username:
standard_user

Password:
secret_sauce

Locked User

Username:
locked_out_user

Password:
secret_sauce

Problem User

Username:
problem_user

Password:
secret_sauce

Performance User

Username:
performance_glitch_user

Password:
secret_sauce

8. Defect Management Process

8.1 Defect Lifecycle

New
→ Assigned
→ In Progress
→ Fixed
→ Retest
→ Closed

OR

New
→ Assigned
→ Rejected

8.2 Defect Severity

Critical

Application crash or checkout failure.

High

Core functionality unavailable.

Medium

Feature partially impacted.

Low

UI or cosmetic issues.

8.3 Defect Priority**P1**

Immediate fix required.

P2

Fix in current release.

P3

Fix in future release.

P4

Minor enhancement.

9. Entry Criteria

Testing will begin when:

- Application URL is available.
- Requirements are understood.
- Test environment is stable.
- Test data is prepared.
- Test cases are reviewed and approved.

10. Exit Criteria

Testing will be considered complete when:

- 100% critical test cases executed.
- All P1 defects closed.
- No open critical defects remain.
- Smoke test suite passes successfully.
- Test summary report is completed.

11. Risks and Mitigation

11.1 Risks and Impact

Risk	Impact	Mitigation
Environment Instability	High	Validate environment before execution
Requirement Ambiguity	Medium	Conduct clarification sessions
Defect retest delays	Medium	Prioritize defect resolution
Browser compatibility issues	Medium	Execute cross-browser testing

11.2 Risk-Based Testing Matrix

A Risk-Based Testing Matrix helps determine which areas require the most testing effort.

11.3 Risk Calculation Formula

Risk Score = Impact × Probability

Value	Description
1	Low
2	Medium
3	High

11.4 Risk Matrix – SauceDemo

Module	Business Impact	Probability of Failure	Risk Score	Risk Level	Testing Focus
Login	3	3	9	Critical	Extensive
Product Inventory	3	2	6	High	Extensive
Shopping Cart	3	3	9	Critical	Extensive
Checkout Information	3	3	9	Critical	Extensive
Checkout Completion	3	3	9	Critical	Extensive
Product Sorting	2	2	4	Medium	Moderate
Product Details	2	2	4	Medium	Moderate
Logout	1	1	1	Low	Basic
Menu Navigation	1	2	2	Low	Basic

11.5 Risk Prioritization

Critical Risk (Score 7-9)

Login
Shopping Cart
Checkout Information
Checkout Completion

Coverage Target: **100%**

Testing Types:

Functional
Negative
Regression
Exploratory

Medium Risk (Score 4-6)

Product Inventory
Product Sorting
Product Details

Coverage Target: 80-90%

Testing Types:

Functional
Regression

Low Risk (Score 1-3)

Logout
Navigation

Coverage Target: **60-70%**

Testing Types:

Smoke
Functional

12. Deliverables

The following deliverables will be produced:

- Test Strategy Document
- Test Plan
- Test Scenarios
- Test Cases
- Defect Reports
- Test Execution Report
- Test Summary Report

13. Test Metrics

The following metrics will be tracked:

- Total Test Cases
- Executed Test Cases
- Passed Test Cases
- Failed Test Cases
- Blocked Test Cases
- Defects Raised
- Defects Closed
- Defect Leakage Rate

14. Approval

Prepared By:

QA Engineer

Reviewed By:

QA Lead

Approved By:

Project Manager